

OTOG QI, China

WMO: 535290

Lat: 39.0883N

Lon: 107.9625E

Elev: 1381

StdP: 85.80

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.9	-18.6	-30.6	0.3	-13.3	-28.3	0.3	-11.9	10.0	-8.5	8.5	-6.5	1.0	20	0.513

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	31.9	15.9	30.6	15.7	29.2	15.4	20.3	25.8	19.4	25.0	18.5	24.2	3.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.9	16.2	22.4	17.8	15.1	21.9	16.6	14.0	21.4	65.4	25.8	61.9	25.2	58.8	24.4	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
9.1	7.6	6.5	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-24.5	34.3	2.7	1.5	-26.5	35.4	-28.1	36.3	-29.6	37.2	-31.6	38.3
			-24.9	21.3	2.6	1.0	-26.8	22.1	-28.3	22.7	-29.8	23.2	-31.7	24.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.1	-8.9	-4.7	2.3	10.1	15.9	20.7	22.9	21.1	15.7	8.6	0.2	-7.1	(6)
(7)		DBStd	11.72	4.29	4.93	5.06	4.74	3.76	2.92	2.56	2.82	3.46	4.09	4.51	4.25	(7)
(8)		HDD10.0	2201	587	413	242	57	4	0	0	3	74	293	529		(8)
(9)		HDD18.3	4074	845	646	496	250	92	11	1	8	92	302	543	787	(9)
(10)		CDD10.0	1522	0	0	4	58	187	322	401	345	173	31	0	0	(10)
(11)		CDD18.3	353	0	0	0	2	18	83	143	94	12	0	0	0	(11)
(12)		CDH23.3	3464	0	0	1	41	263	878	1354	794	130	3	0	0	(12)
(13)		CDH26.7	1086	0	0	0	7	50	265	511	226	27	0	0	0	(13)
(14)	Wind	WSAvg	2.6	2.2	2.5	3.0	3.2	3.2	2.8	2.6	2.4	2.3	2.3	2.5	2.3	(14)
(15)	Precipitation	PrecAvg	262	2	2	7	11	22	25	60	81	37	13	4	1	(15)
(16)		PrecMax	446	9	9	28	66	84	63	153	249	128	50	20	10	(16)
(17)		PrecMin	125	0	0	0	0	0	3	16	7	0	1	0	0	(17)
(18)		PrecStd	94	2	2	7	14	22	15	34	63	30	10	6	2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.6	11.4	20.3	27.5	30.2	32.9	34.7	32.2	30.1	23.4	15.1	6.7	(19)
MCWB			-0.9	2.1	7.5	11.2	12.3	14.3	17.8	16.7	13.1	10.8	5.3	-0.4	(20)	
2%		DB	3.4	8.1	16.4	24.0	28.0	31.0	32.6	30.7	26.4	20.8	12.6	4.1	(21)	
		MCWB	-2.5	0.4	5.3	9.2	11.7	14.1	16.7	16.7	13.6	10.0	4.1	-1.7	(22)	
5%		DB	0.9	5.8	13.6	21.5	26.4	29.5	31.1	29.2	24.5	18.5	10.3	2.1	(23)	
		MCWB	-4.1	-0.8	3.8	8.3	11.3	13.9	16.6	16.7	13.2	8.8	2.9	-2.9	(24)	
10%		DB	-1.3	3.7	11.2	19.1	24.5	28.1	29.7	27.7	22.7	16.3	8.0	0.0	(25)	
		MCWB	-5.4	-1.8	2.5	7.2	10.4	13.7	16.4	16.2	12.9	7.8	1.9	-4.0	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.5	3.0	8.4	12.8	15.1	18.7	21.7	21.5	19.0	13.1	7.0	0.3	(27)
MCDWB			6.1	9.8	17.5	20.0	22.3	24.0	27.0	26.7	21.6	19.6	12.3	5.4	(28)	
2%		WB	-2.4	1.2	6.0	11.1	13.7	17.2	20.5	20.4	16.8	11.6	5.2	-1.4	(29)	
		MCDWB	3.0	6.8	14.7	19.6	22.6	23.4	26.3	25.4	20.8	17.6	10.1	3.4	(30)	
5%		WB	-3.8	-0.3	4.4	9.7	12.6	16.3	19.6	19.5	15.5	10.3	3.8	-2.6	(31)	
		MCDWB	0.6	5.0	12.6	18.5	22.0	23.0	25.9	24.7	20.6	15.8	8.8	1.7	(32)	
10%		WB	-5.3	-1.5	3.0	8.3	11.8	15.5	18.8	18.6	14.4	9.1	2.4	-3.9	(33)	
		MCDWB	-1.4	3.6	10.4	17.7	21.0	23.0	24.9	23.9	19.9	14.2	7.7	0.0	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	12.6	12.9	12.7	12.9	12.7	12.0	11.0	10.8	11.2	12.1	11.7	11.5	(35)
MCDBR			15.4	15.3	15.9	16.3	15.7	14.5	13.6	13.1	14.1	15.0	14.4	13.7	(36)	
MCWBR			10.8	9.7	8.8	7.7	6.1	4.4	3.9	4.0	5.1	6.8	8.1	9.4	(37)	
5% WB		MCDBR	14.7	14.0	14.2	13.2	12.6	10.8	9.9	9.6	9.7	12.3	12.4	12.8	(38)	
		MCWBR	10.4	9.1	8.1	6.7	5.7	3.8	4.0	4.1	4.8	6.5	7.7	8.9	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.323	0.380	0.424	0.468	0.493	0.521	0.492	0.495	0.471	0.419	0.349	0.315	(40)	
taud		2.118	1.980	1.897	1.833	1.839	1.837	1.990	1.959	1.944	1.998	2.106	2.164	(41)		
Ebn at Noon		822	814	819	809	797	774	793	778	766	761	780	801	(42)		
Edh at Noon		111	148	178	203	206	207	176	177	169	143	112	98	(43)		
(44)	All-Sky Solar Radiation	RadAvg	2.75	3.69	5.02	6.06	6.76	6.54	6.16	5.59	4.67	4.02	3.03	2.51	(44)	
(45)		RadStd	0.28	0.24	0.32	0.41	0.34	0.47	0.27	0.27	0.33	0.25	0.28	0.20	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	-1.13	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page